

U-space USSP-USSP Exchanges for UAS Flight Authorisation SWIM Service Description Document

General Service Description	
Service description identification SWIM-SERV-030	Title: U-space USSP-USSP Exchanges for UAS Flight Authorisation SWIM Service Description Edition: V1.0 Reference Date: 2025-11-01
Service identification SWIM-SERV-040	Name: U-space USSP-USSP Exchanges for UAS Flight Authorisation Service Version: V1.0
Service abstract SWIM-SERV-050	The UAS Flight Authorisation service is one of the four mandatory services under the Commission Implementing Regulation (EU) 2021/664. The UAS flight authorisation service provides authorisations to UAS operators for each individual flight based on other notified flight requests that may conflict with other unmanned operations within the same U-space airspace. It is a strategic deconfliction tool. This service implementation is based on the ASTM F3548-21 standard, identified as an applicable standard in U-space accepted means of compliance and guidance material.
Service Provider and Contact Information	
Service Provider SWIM-SERV-060	Name: U-space Service Provider (see Operations Committee list for participating providers) Description of organization: U-space Service Provider refers to an organisation that is certified by a competent authority to provide U-space services in the U-space airspace (Ref: GM1a Article 7 U-space service providers)
Service Characteristics	
Geographical extent of information SWIM-SERV-090	Geographical coverage of exchanged information service payload is available through the DSS under ASTM Section A2.

Service categories SWIM-SERV-100	Service category: Flight information
Service lifecycle information SWIM-SERV-110	Lifecycle stage and if applicable qualifying dates: As published by the Operations Committee.
Service standard reference SWIM-SERV-120	Service standard reference: Standard Specification for UAS Traffic Management (UTM) UAS Service Supplier (USS) Interoperability (ASTM F3548-21) Implemented options of the service standard: • Priority level for all operational intents will be set to 0 • Per U-space regulation Article 10, conflicts are not permitted within the same priority level • Requirements in Section A2.4 regarding down USS not used Discovery and synchronization service configuration process - as defined by InterUSS DSS Pooling instructions on joining an existing pool with new instance.
High-level Description of Service Offer	
Operational environment SWIM-SERV-130	Refer to the following documents for operational needs addressed by the service and the capabilities offered by the service. • Commission Implementing Regulation (EU) 2021/664 - Preamble • ASTM F3548-21 Section A2 capabilities include: - USS-DSS Interfaces - Other DSS Interfaces - USS-USS Interfaces - DSS-DSS Synchronization
Service functions SWIM-SERV-140	Refer to the following documents for functions offered by the service and their real world effects. • Commission Implementing Regulation (EU) 2021/664 Article 10 • Article 10 Accepted Means of Compliance and Guidance Material to Regulation (EU) 2021/664 — Issue 1 • ASTM F3548-21 Section A2 functions are provided for the following capabilities: - USS-DSS Interfaces - Other DSS Interfaces - USS-USS Interfaces - DSS-DSS Synchronization
Limitations and Constraints on Using the Service	
Service access and	Refer to the EU USSP U-space Data Sharing and Governance

use conditions SWIM-SERV-150	Agreement for information on legal constraints, service policies, and service consumption constraints Service operational policies: With reference to Regulation (EU) 2021/664: * • USSPs shall use 4D volumes exchanged as a part of this service to detect conflicts among flight authorisation requests. The USSP-Operator agreement will ensure that the 4D volumes exchanged among USSPs consider U-space requirements such as inclusion of deviation thresholds and contingency/emergency measures as applicable • For the purposes of V1, deviation thresholds are set to be 0 • * A variant 1 (RFC 4122), Version 4 UUID shall be used for all shared flight authorisations.
Security constraints SWIM-SERV-160	. For field testing (U-space v1), security compliance shall be demonstrated either by: ○ holding a valid USSP certification in accordance with Regulation (EU) 2021/664, or ○ holding an ISO/IEC 27001 certification and being in the process of obtaining USSP certification, or ○ complying to comparable security industry standards (such as IEC 62443) and being in the process of obtaining USSP certification. This recognition is a transitional measure limited to field testing and is not applicable to full U-space deployment. ASTM F3548-21 Sections 5.2 and A2 (relevant requirements are listed below): * GEN0005 * DSS0200 * DSS0205 <i>Per ASTM F3548-21, Appendix X1 - Reference architecture for interoperability security controls, The X1.1 Base Deployment: Access Tokens with Audience Claims is implemented. Authentication and Authorization for communication between USS-USS and USS-DSS shall be facilitated by a common interoperability token exchanger service supporting OAuth 2.0.</i>
Quality Aspects	
Quality of service SWIM-SERV-180	Refer to the following documents for quality requirements. • EU USSP U-space Data Sharing and Governance Agreement - Annex XX Service Level Agreement for information on quality of service

Service validation information SWIM-SERV-200	Service will be verified using the InterUSS Automated Test Baseline prior to software deployment using test baseline identifier TB-1e05e87 .
Behavior of the Service	
Application message exchange pattern SWIM-SERV-210	Refer to the following sections in ASTM F3548-21 for information about the application message exchange pattern(s) used by the service. * A2 - Interoperability requirements and DSS testing * A3 - USS-DSS and USS-USS OpenAPI YAML Description
Service behavior SWIM-SERV-220	Refer to the attached Compliance Matrix, <i>U-space Article 10 UAS Flight Authorisation Compliance Matrix V1.0</i> for service performance and verification requirements
Service Implementation and Structural Details	
Service interfaces SWIM-SERV-240	Refer to the following sections in ASTM F3548-21 for information about service interfaces. * A2 - Interoperability requirements and DSS testing * A3 - USS-DSS and USS-USS OpenAPI YAML Description
Service interface protocols and data format SWIM-SERV-260	Refer to the following sections in ASTM F3548-21 for information about service interface protocols and data format. * A2 - Interoperability requirements and DSS testing * A3 - USS-DSS and USS-USS OpenAPI YAML Description
Service operations SWIM-SERV-270	Refer to the following sections in ASTM F3548-21 for information about service operations. * A2 - Interoperability requirements and DSS testing * A3 - USS-DSS and USS-USS OpenAPI YAML Description
Service messages SWIM-SERV-280	Refer to the following sections in ASTM F3548-21 for information about messages exchanged by the service. * A2 - Interoperability requirements and DSS testing * A3 - USS-DSS and USS-USS OpenAPI YAML Description
Information Aspects of the Service	
Information definition (minimum) SWIM-SERV-290	Refer to the following sections in ASTM F3548-21 for information about the exchanged information service payload. * A2 - Interoperability requirements and DSS testing * A3 - USS-DSS and USS-USS OpenAPI YAML Description
Resources	

Machine-readable service interface definition SWIM-SERV-320	Refer to the following sections in ASTM F3548-21 for information about the exchanged information service payload. * A2 - Interoperability requirements and DSS testing * A3 - USS-DSS and USS-USS OpenAPI YAML Description
Abbreviations and acronyms SWIM-SERV-350	Refer to the following sections in ASTM F3548-21 for abbreviations and acronyms: * 3 - Terminology * A2 - Interoperability requirements and DSS testing * A3 - USS-DSS and USS-USS OpenAPI YAML Description